178052: biotite microgranodiorite, Yuinmery Homestead

Location and sampling

YOUANMI (SH 50-4), RAYS ROCKS (2740)
MGA Zone 50, 697330E 6843460N

Sampled on 29 October 2002

The sample was taken from a tor about 2 m high and 8 m in diameter 30 m east of the road from Sandstone to Yuinmery Homestead and 4.6 km north of Yuinmery Homestead.

Tectonic unit/relations

The sample is from a pale-grey, fine- and even-grained, undeformed biotite granodiorite 1.5 km east of the Yuinmery Shear Zone, Southern Cross Granite–Greenstone Terrane, Yilgarn Craton (Chen, 2002). It was collected to determine the age of undeformed rocks near the shear zone.

Petrographic description

The principal minerals are plagioclase (60 vol.%), quartz (20 vol.%), K-feldspar (15 vol.%), biotite (4 vol.%), opaque oxide (≤1 vol.%), and titanite (≤1 vol.%), with trace amounts of accessory apatite, chlorite, epidote, ?allanite, sericite, clinozoisite, carbonate, and zircon. Plagioclase forms laths up to 3 mm long, possibly defining a weak flow foliation in some areas. Normal zoning is present, apparently from andesine to oligoclase, with rims having less than about 15 vol.% anorthite. There is minor interstitial microcline as poikilitic grains up to 4 mm in diameter, and more-abundant interstitial quartz. Some quartz is euhedral against the microcline, with suggestions of a bipyramidal habit implying a shallow emplacement level. Minor biotite, up to 1 mm in grain size, is unoriented and accompanied by fine granular oxide and mostly euhedral titanite crystals less than 1 mm long. Apatite is a common accessory mineral. There are grains of epidote with altered cores that could have been allanite. The plagioclase has patches and zones altered to sericite and clinozoisite, locally with carbonate. Alteration of biotite to chlorite is weak, but minor epidote has developed in and adjacent to biotite in some areas. In addition, some of the titanite is altered to leucoxene, but alteration is weak and of low temperature origin. The fabric shows a lack of deformation, suggesting a shallow emplacement level.

Zircon morphology

The zircons isolated from this sample are typically colourless to pale greenish-yellow, dark brown or black, euhedral and elongate whole grains and grain fragments, and between 50 × 80 µm and 160 × 80 µm in size. Most grains are internally structureless, but some have faint internal zonation. Cathodoluminescence images of representative zircons are given in Figure 1.

Analytical details

This sample was analysed on 23 October 2003. The counter deadtime during the analysis session was 24 ns. Seven analyses of the CZ3 standard obtained during the analysis session indicated a Pb*/U calibration uncertainty of 0.999% (1σ). A calibration uncertainty of 1.0% (1σ) was applied to analyses of unknowns obtained during this analysis session. Common-Pb corrections were applied assuming Broken Hill common-Pb isotopic compositions for all analyses, with the exception of analyses 6.1 and 3.2, for which isotopic compositions determined using the method of Cumming and Richards (1975) were assumed.

Results

Seventeen analyses were obtained from nine zircons. Results are given in Table 1 and shown on a concordia plot in Figure 2.

Interpretation

The analyses are concordant to highly discordant, with the discordance pattern consistent with a dominant recent episode of radiogenic-Pb loss. Sixteen concordant and highly discordant analyses of nine zircons have $^{207}Pb/^{206}Pb$ ratios defining a single population and indicating a weighted mean $^{207}Pb/^{206}Pb$ date of 2688 ± 11 Ma (chi-squared = 1.41). Concordant analysis 2.2 indicates a slightly younger $^{207}Pb/^{206}Pb$ date than the main population.

The date of 2688 ± 11 Ma indicated by the weighted mean $^{207}Pb/^{206}Pb$ ratio of 16 concordant and highly discordant analyses of nine zircons is interpreted as the age of igneous crystallization of the microgranodiorite. Concordant analysis 2.2 is interpreted to be of a site that has undergone ancient radiogenic-Pb loss.

Recommended reference for this publication:

NELSON, D. R., 2005, 178052: biotite microgranodiorite, Yuinmery Homestead; Geochronology dataset 581; in Compilation of geochronology data, June 2006 update: Western Australia Geological Survey.

Data obtained: 23/10/2003; Data released: 30/06/2005

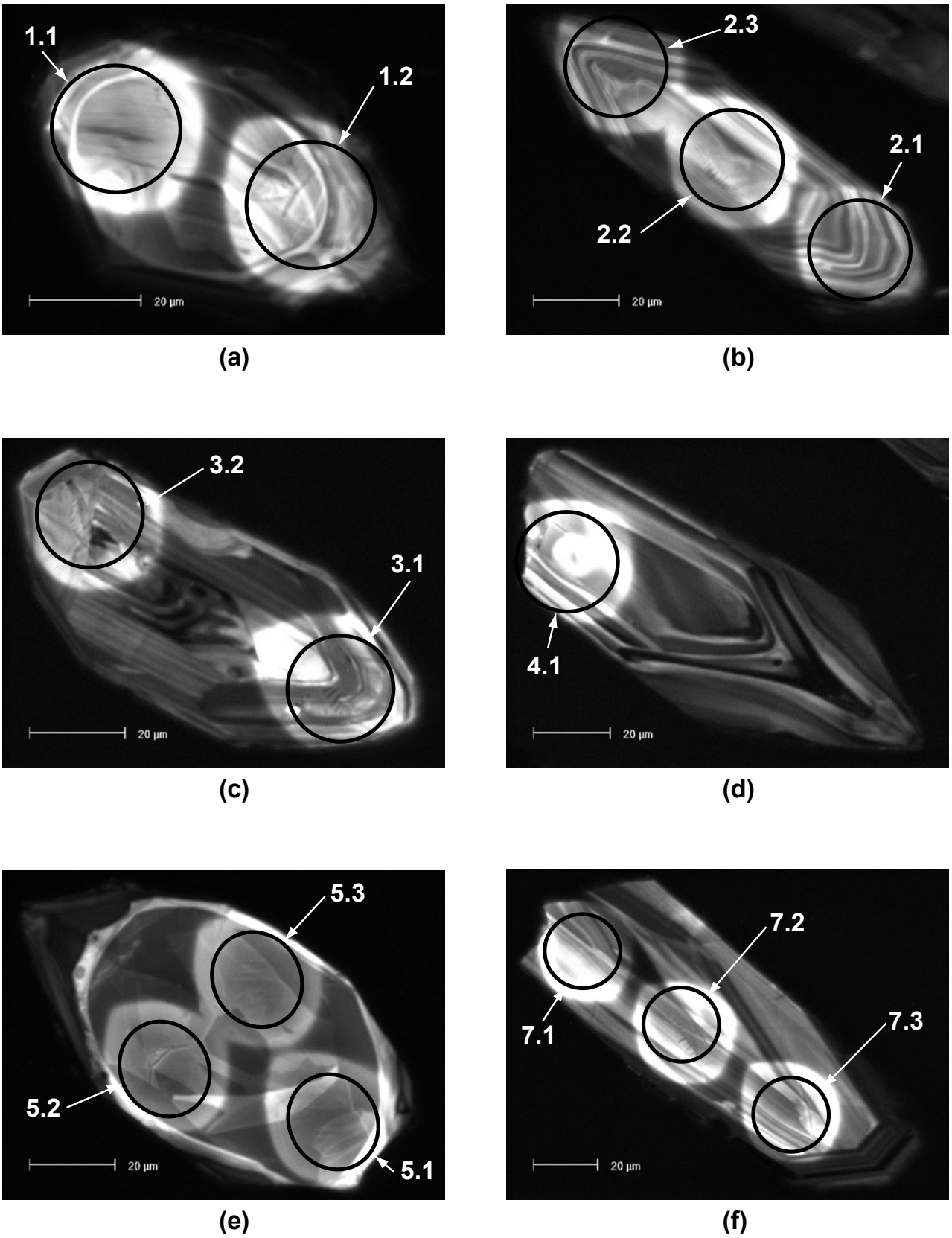

Figure 1. Cathodoluminescence images of representative zircons from sample 178052: biotite microgranodiorite, Yuinmery Homestead

Table 1. Ion microprobe analytical results for zircons from sample 178052: biotite microgranodiorite, Yuinmery Homestead

Grain .spot	*U (ppm)*	*Th (ppm)*	*Pb (ppm)*	*f206%*	$^{207}Pb/^{206}Pb$	±1σ	$^{208}Pb/^{206}Pb$	±1σ	$^{206}Pb/^{238}U$	±1σ	$^{207}Pb/^{235}U$	±1σ	*% concordance*	$^{207}Pb/^{206}Pb$ *Age*	±1σ
1.1	40	44	27	0.601	0.19119	0.00293	0.30831	0.00614	0.5139	0.0066	13.548	0.288	97	2 752	25
2.1	120	77	67	1.971	0.18445	0.00212	0.19878	0.00439	0.4421	0.0049	11.244	0.190	88	2 693	19
3.1	224	280	149	0.802	0.18437	0.00109	0.35185	0.00233	0.4908	0.0051	12.476	0.158	96	2 693	10
4.1	112	104	73	1.577	0.18194	0.00192	0.25839	0.00403	0.5026	0.0056	12.607	0.205	98	2 671	17
5.1	88	94	60	0.427	0.18647	0.00157	0.29599	0.00312	0.5236	0.0059	13.462	0.201	100	2 711	14
6.1	201	155	110	6.542	0.18402	0.00308	0.29813	0.00676	0.3501	0.0038	8.883	0.187	72	2 689	28
7.1	94	69	47	4.140	0.18496	0.00352	0.25917	0.00773	0.3594	0.0042	9.165	0.216	73	2 698	31
8.1	109	100	71	0.611	0.18592	0.00158	0.24882	0.00309	0.5121	0.0057	13.127	0.194	98	2 706	14
9.1	128	152	81	2.106	0.18147	0.00210	0.33526	0.00463	0.4565	0.0050	11.423	0.194	91	2 666	19
1.2	43	40	27	2.735	0.18440	0.00398	0.25943	0.00863	0.4717	0.0061	11.992	0.318	92	2 693	36
2.2	59	46	36	1.636	0.17464	0.00268	0.19596	0.00551	0.4913	0.0059	11.831	0.243	99	2 603	26
2.3	169	110	88	1.067	0.18414	0.00161	0.21411	0.00320	0.4188	0.0045	10.634	0.156	84	2 691	14
3.2	162	142	106	4.917	0.17991	0.00275	0.27605	0.00600	0.4502	0.0049	11.168	0.222	90	2 652	25
5.2	90	129	65	0.710	0.18290	0.00187	0.39156	0.00416	0.5175	0.0059	13.050	0.213	100	2 679	17
5.3	89	101	60	0.625	0.18405	0.00179	0.30623	0.00366	0.5170	0.0059	13.121	0.209	100	2 690	16
7.2	80	79	50	1.341	0.17948	0.00219	0.25218	0.00454	0.4816	0.0056	11.918	0.212	96	2 648	20
7.3	78	69	50	0.925	0.18058	0.00208	0.24336	0.00416	0.5036	0.0058	12.539	0.217	99	2 658	19

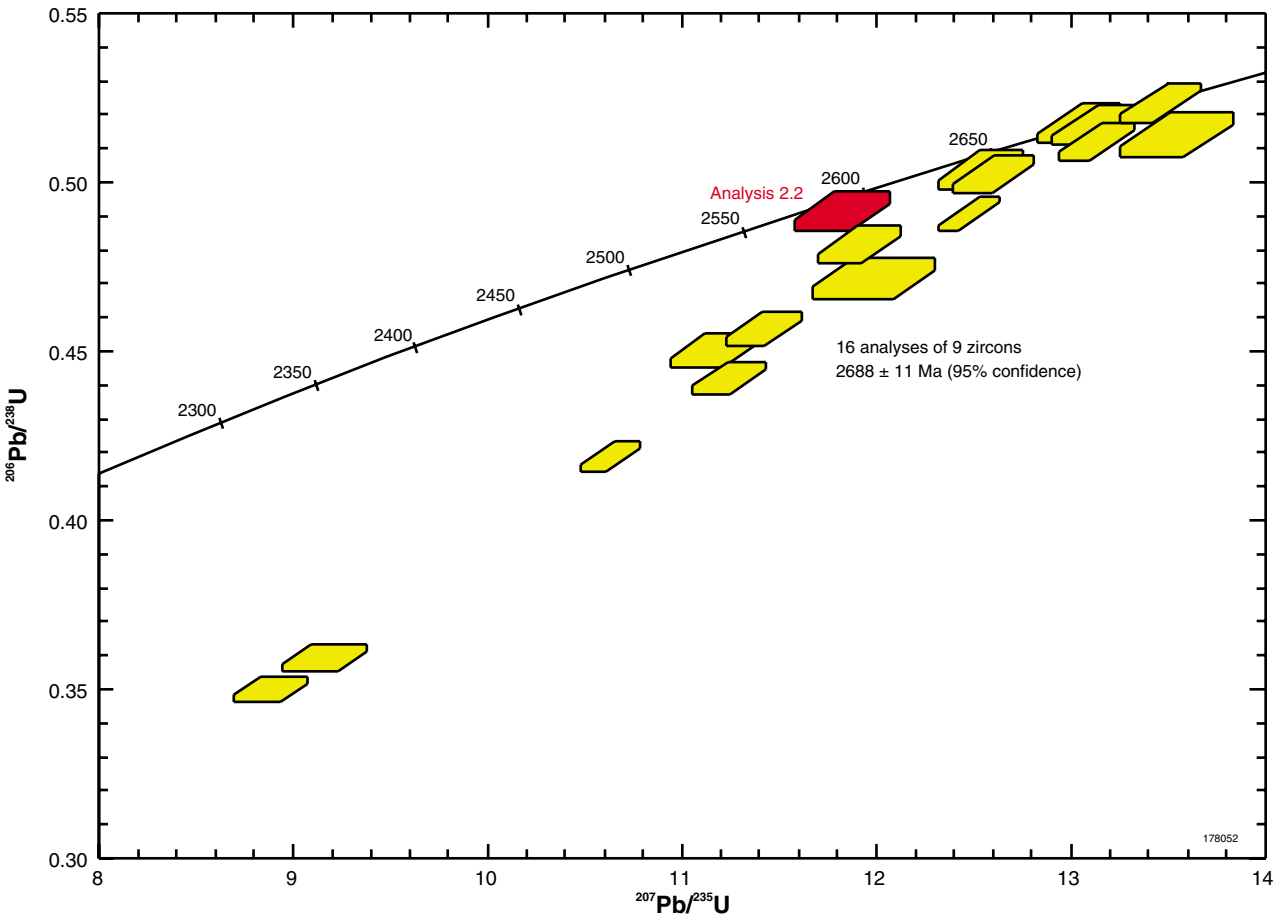

Figure 2. Concordia plot for sample 178052: biotite microgranodiorite, Yuinmery Homestead